



**JERUSALEM COLLEGE OF ENGINEERING, CHENNAI – 600 100**  
**(An Autonomous Institution Affiliated to Anna University, Chennai)**

Reg. No. :

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**Question Paper Code : 147446**

B.E. DEGREE END SEMESTER EXAMINATIONS, NOV/DEC 2025 & JAN 2026

Sixth Semester

Computer Science and Engineering (Artificial Intelligence and Machine Learning)

JAL1603 – TIME SERIES ANALYSIS AND FORECASTING

(Regulation 2021)

Time : Three hours

Maximum : 100 marks

Answer ALL questions

PART A – ( $10 \times 2 = 20$  marks)

1. What does run sequence plot indicate?
2. List the uses of autoregressive models.
3. Write short notes on auto covariance function.
4. Define variogram. Give an example.
5. What is the test for significance of regression? Mention the hypotheses for the same.
6. Define influential points.
7. Write short notes on infinite moving average.
8. How do ACF and PACF help in AR and MA models?
9. What are the three main layers in a typical feedforward artificial neural network?
10. What is the Bayes' estimator of  $\theta$ ?

PART B – ( $5 \times 16 = 80$  marks)

11. (a) What are the different types of data used in analytics? Explain their purpose with suitable examples.

Or

11. (b) List down and explain each of the activities in the forecasting process.

12. (a) Describe the basic steps in modelling and forecasting a time series.

Or

12. (b) Explain about the different ways of selecting an appropriate forecasting model.

13. (a) Explain how residual plots and scaled residuals support in checking the adequacy of a model.

Or

13. (b) What is the purpose of Durbin Watson test and Cochrane-Orcutt method? Explain in detail.

14. (a) What are ARMA models? Explain ARMA(p, q) processes.

Or

14. (b) Explain the procedure for building time series models.

15. (a) Explain in detail about how neural networks can be used forecasting .

Or

15. (b) Explain in detail how Bayesian methods can be used in forecasting.

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